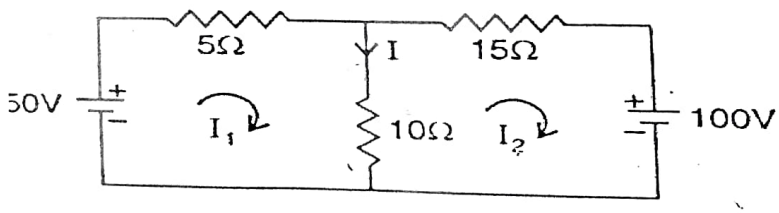
(Please write your Enrollment Number) Enrollment No. 02701182025

End-Term Examination
(Non-CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech. Semester: 1st
(December, 2025)

Subject Code: BEC 105	**Subject: Fundamentals of Electrical Sciences**
Time :3 Hours	**Maximum Marks :60**

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

		(2.5*8=20)
Q1	a) Explain Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) with suitable examples.	
	b) Differentiate between Mesh Analysis and Nodal Analysis with suitable examples	
	c) Explain the Superposition Theorem and its application in solving DC circuits.	
	d) Draw the phasor diagram for a series RL circuit and derive the expression for impedance.	
	e) Define voltage regulation in a single-phase transformer.	
	f) What is resonance in AC circuits? Briefly explain series resonance.	
	g) Describe the principle of operation of a moving coil ammeter.	
	h) Explain the constructional features of a DC generator.	
	UNIT I	
Q2	Find out the current I flowing through the 10 ohm resistor shown in Fig- 1 using (a). Mesh Analysis method (b). Nodal Analysis method.  Fig: 1	(10)
Q3	(a). State and explain Millman's Theorem (b). Using Thevenin's Theorem, find out the current I flowing through the 10 ohm resistor shown in Fig-1	(10)
	UNIT II	
Q4	For a series RLC circuit, derive the expressions for resonant frequency, quality factor (Q-factor), and bandwidth. Plot the frequency response curve.	(10)
Q5	Explain the two-wattmeter method for measuring power in a three-phase system. Derive the expressions for total power and power factor.	(10)
	UNIT III	
Q6	Describe the construction and working principle of a moving iron instrument. Compare it with a moving coil instrument in terms of accuracy and applications.	(10)

Q7	Discuss the principle and construction of a dynamometer type wattmeter. Explain its errors and methods to minimize them.	(10)
	UNIT IV	
Q8	Draw the equivalent circuit of a single-phase transformer and derive the EMF equation. Explain core losses and copper losses.	(10)
Q9	Explain the construction and working principle of a DC motor. Derive the back EMF equation and torque equation.	(10)